

OLIST E-COMMERCE SQL BUSINESS CASE STUDY

Business Questions • SQL Analysis • Actual Results • Insights • Recommendations

A manager/client-style business insights report built from the SQL portfolio analysis.

99,441	96,096	\$16,008,872.12	\$160.99
orders	unique customers	payment value	average order value

Dataset period: 04 Sep 2016 – 17 Oct 2018. Currency displayed using \$ for portfolio presentation.

1. Executive Summary

This report translates the SQL portfolio into a concise set of management-level findings. The focus is on what the analysis means for the business, not on reproducing every SQL query.

BUSINESS QUESTION	SQL ANALYSIS
What are the most important commercial and operational signals in the dataset?	Combine the core SQL outputs for orders, customers, revenue, AOV, delivery, payments and reviews.
ACTUAL RESULT	BUSINESS INSIGHT
99,441 orders • 96,096 unique customers • 3,095 sellers • 32,951 products • 73 categories. AOV: \$160.99 • Delivered orders: 97.0% • Avg delivery: 12.5 days .	Revenue is concentrated geographically and by leading categories. Delivery is a meaningful customer-experience lever: delayed orders score 2.57/5 vs 4.29/5 on-time.
RECOMMENDATION	Prioritize delivery reliability, core-market performance, high-value customers and leading categories as recurring management KPIs.

Headline findings

- **SP** contributes 37.5% of payment revenue.
- **Credit Card** contributes 78.3% of payment value.
- **Health Beauty** is the leading category by item-price revenue at \$1,258,681.34.
- The strongest revenue month is **2017-11** at \$1,194,882.80.

2. Business Overview & Commercial Performance

BUSINESS QUESTION	SQL ANALYSIS
How large is the Olist marketplace and what is its average order value?	COUNT(DISTINCT) across customers, sellers, products, categories and orders; aggregate payment_value by order_id and calculate AVG(order_value).
ACTUAL RESULT	BUSINESS INSIGHT
99,441 orders, 96,096 unique customers, 3,095 sellers, 32,951 products and 73 categories. Average order value: \$160.99 .	The marketplace has enough scale for customer, seller, product and geographic segmentation. AOV provides a baseline for evaluating commercial performance.
RECOMMENDATION	Track orders, payment value and AOV together in recurring management reporting.

Why this matters

A strong business overview establishes the scale of the marketplace before drilling into customer, seller, product or operational performance.

3. Product & Category Insights

BUSINESS QUESTION	SQL ANALYSIS
Which product categories generate the highest revenue and volume?	JOIN order_items to products and translated categories; SUM(price) by category; COUNT(order_item_id) by category.
ACTUAL RESULT	BUSINESS INSIGHT
Top revenue categories: 1. Health Beauty — \$1,258,681.34 2. Watches Gifts — \$1,205,005.68 3. Bed Bath Table — \$1,036,988.68 4. Sports Leisure — \$988,048.97 5. Computers Accessories — \$911,954.32 Top unit-volume category: Bed Bath Table (11,115 units).	Revenue concentration and unit concentration are not identical. A category can lead in volume without leading in monetary value.
RECOMMENDATION	Evaluate categories on both revenue and unit volume; protect inventory and seller coverage in the strongest categories.

4. Customer & Revenue Insights

BUSINESS QUESTION	SQL ANALYSIS
Which customers have the highest cumulative spending, and how different are customer value segments?	JOIN customers → orders → order_payments; SUM(payment_value) by customer_unique_id; rank customers and divide spending into four groups.
ACTUAL RESULT	BUSINESS INSIGHT
Top customer spending: 1. Customer 0a0a92112bd4c708ca5fde585afaa872 — \$13,664.08 2. Customer 46450c74a0d8c5ca9395da1daac6c120 — \$9,553.02 3. Customer da122df9eeddfedc1dc1f5349a1a690c — \$7,571.63 4. Customer 763c8b1c9c68a0229c42c9fc6f662b93 — \$7,274.88 5. Customer dc4802a71eae9be1dd28f5d788ceb526 — \$6,929.31 Average spending rises from \$43.66 in Q1 to \$397.48 in Q4.	Customer value is highly uneven. A relatively small group contributes much more spending than lower-value customers.
RECOMMENDATION	Create differentiated retention, loyalty and cross-sell strategies by customer spending segment.

5. Geographic Revenue Insights

BUSINESS QUESTION	SQL ANALYSIS
Which customer states generate the highest sales revenue?	JOIN customers → orders → order_payments; SUM(payment_value) by customer_state; calculate revenue share.
ACTUAL RESULT	BUSINESS INSIGHT
1. SP — \$5,998,226.96 (37.5%) 2. RJ — \$2,144,379.69 (13.4%) 3. MG — \$1,872,257.26 (11.7%) 4. RS — \$890,898.54 (5.6%) 5. PR — \$811,156.38 (5.1%) Top 3 states combined: 62.6%.	Revenue is geographically concentrated, so service quality and commercial execution in SP, RJ and MG have an outsized impact.
RECOMMENDATION	Protect the core markets while using AOV, delivery and satisfaction metrics to identify expansion opportunities.

6. Seller Insights

BUSINESS QUESTION	SQL ANALYSIS
Which sellers generate the highest revenue and how can sellers be benchmarked fairly?	SUM(order_items.price) by seller_id; use DENSE_RANK() with PARTITION BY seller_state for within-state ranking.
ACTUAL RESULT	BUSINESS INSIGHT
Top global sellers by item-price revenue: 1. 4869f7a5dfa277a7dca6462dcf3b52b2 — \$229,472.63 2. 53243585a1d6dc2643021fd1853d8905 — \$222,776.05 3. 4a3ca9315b744ce9f8e9374361493884 — \$200,472.92 4. fa1c13f2614d7b5c4749cbc52fecda94 — \$194,042.03 5. 7c67e1448b00f6e969d365cea6b010ab — \$187,923.89 The project also ranks sellers within each seller state.	Seller revenue is concentrated, but revenue alone does not capture service quality. Within-state ranking provides a fairer peer comparison.
RECOMMENDATION	Build a seller scorecard combining revenue, order volume, review score, delivery time and freight performance.

7. Payment Insights

BUSINESS QUESTION	SQL ANALYSIS
Which payment methods generate the most payment value?	SUM(payment_value) by payment_type and calculate each method's share of total payment value.
ACTUAL RESULT	BUSINESS INSIGHT
1. Credit Card — \$12,542,084.19 (78.3%) 2. Boleto — \$2,869,361.27 (17.9%) 3. Voucher — \$379,436.87 (2.4%) 4. Debit Card — \$217,989.79 (1.4%) 5. Not Defined — \$0.00 (0.0%)	Credit card is the dominant payment method, accounting for the large majority of payment value.
RECOMMENDATION	Maintain a frictionless credit-card checkout while monitoring alternative payment methods by customer segment.

8. Delivery & Customer Satisfaction Insights

BUSINESS QUESTION	SQL ANALYSIS
How does delivery performance affect customer satisfaction?	Calculate delivery days for delivered orders; compare review_score for Delayed vs On-Time orders.
ACTUAL RESULT	BUSINESS INSIGHT
Average delivery time: 12.5 days . Delayed: 2.57/5 • On-Time: 4.29/5 • Review gap: 1.73 points . Delayed share: 8.0% .	Delivery reliability is strongly associated with customer satisfaction in this dataset. It is a customer-experience issue, not merely a logistics KPI.
RECOMMENDATION	Prioritize delayed-order reduction and monitor delivery SLA and review score together.

Geographic operational signal

The five slowest customer states by average delivery time are:

RR — 29.3 days

AP — 27.2 days

AM — 26.4 days

AL — 24.5 days

PA — 23.7 days

9. Time & Growth Insights

BUSINESS QUESTION	SQL ANALYSIS
Which months generated the strongest revenue and growth?	Group payment value by purchase month; use LAG() to compare the current month with the previous month.
ACTUAL RESULT	BUSINESS INSIGHT
Highest revenue month: 2017-11 — \$1,194,882.80 . Strongest MoM percentage increase: 2017-01 — 705751.4% .	Absolute revenue shows scale, while MoM growth shows acceleration or slowdown. Both are useful management signals.
RECOMMENDATION	Track monthly revenue, order volume, AOV and MoM growth together and annotate unusual changes.

10. Advanced SQL & Analytical Applications

Business Question	SQL Technique	Business Use
Latest order for each customer	ROW_NUMBER()	Identify recent customer activity.
Top seller within each state	DENSE_RANK() + PARTITION BY	Fair regional benchmarking.
Revenue change from previous month	LAG()	Period-over-period growth.
Cumulative revenue	SUM() OVER(ORDER BY ...)	Track accumulated revenue.
Customer revenue contribution	SUM() OVER()	Measure customer share of total.
Customer spending segments	NTILE(4)	Create actionable value segments.
Peer benchmarking by state	AVG() OVER(PARTITION BY ...)	Compare customers against local averages.

Portfolio value

These window-function analyses demonstrate how the project moves beyond basic aggregation into customer recency, ranking, segmentation, benchmarking and time-series analysis.

11. Management Recommendations

1

Improve delivery reliability

Delayed orders are associated with a 1.73-point lower average review score.

2

Protect core markets

SP, RJ and MG together contribute 62.6% of payment revenue.

3

Focus category investment

Support leading categories with inventory, seller coverage and targeted promotions.

4

Build a customer value strategy

Use spending segments for differentiated retention, loyalty and cross-sell programs.

5

Create a seller scorecard

Combine revenue with reviews, delivery time, freight and order volume.

6

Operationalize the SQL

Turn revenue, orders, AOV, delay rate, review score and repeat behavior into recurring KPIs.